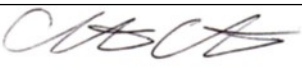


Guideline Title: End-Tidal CO2 Monitoring

Guideline Number: GUID-3203-PR009

Issue date: 09/09/2014	Replaces Dept. Policy: End-Tidal CO2 Detector
Revision dates: 4/30/24	Developed by: Air Care Team
Approved by: Dr. Christopher Hunter, MD Air Care Team Medical Director	Approved by:
Signature: 	Signature:
Department Numbers: 3203	

- I. **PURPOSE:** To assist in the confirmation of proper endotracheal tube placement
- II. **DEFINITIONS:**
 - a. ETT- Endotracheal tube
 - b. ETCO2 – End tidal carbon dioxide
 - c. GCS – Glasgow Coma Scale
- III. **DEPARTMENT GUIDELINE:**
 - a. **EQUIPMENT:**
 - i. In-line ETCO2 detector
 - ii. Nasal cannula ETCO2 detector
 - iii. Monitor with digital capnometry and capnography cable/sensor/adaptor.
 - b. **INDICATIONS:**
 - i. In line: All patients who are intubated, have a supraglottic airway device [King tube, laryngeal mask airway (LMA), I Gel] or advanced alternative airway in place.
 - ii. Nasal Cannula Capnometry
 1. Use in non-intubated patients with a GCS < 14 who receive sedation or analgesia.
 2. Patients who are a GCS > 8 but show signs of decreased work of breathing.
 - c. **CONTRAINDICATIONS:** None.
 - d. **PROCEDURE:**
 - i. Digital Capnometry or waveform Capnography are the referenced standards for confirmation and maintenance of ETT placement.
 - ii. **Digital Capnometry and Capnography:**
 1. Choose age/size appropriate adapter for specific monitor.
 2. Prepare monitor for use.
 3. Disconnect the tube from the ventilation circuit and connect the ETCO2 adapter to the tube or if High Efficiency Particulate Air (HEPA) filter present, connect between the filter and the ventilator.
 4. Reconnect the ventilation circuit and resume ventilation of the patient.
 5. Observe the unit for a numeric value and a waveform.

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- a. An appropriate, reproducible, 4-phase waveform with a numeric value confirms that the tube is placed in the trachea.
 - b. An inappropriate, flatline, or non-reproducible waveform suggests the tube is incorrectly placed. Remove the ETT and replace with a secure airway.
 - c. A reproducible, 4-phase waveform with an extremely low (≤ 10 mmHg) ETCO2 suggests very poor perfusion or extreme acidosis.
 6. Begin continuous monitoring of capnometric and capnographic ETCO2 to give a breath-by-breath status of tube placement and ventilatory adequacy throughout transport.
 7. ETCO2 goal
 - a. 35-45 mmHg for most patients
 - b. 35-40 mmHg if head injury present
 - c. 30-35 mmHg if there is evidence of herniation with head injury per GUID3203-T014 – *Traumatic Brain Injury*.
 8. Assess waveform for bronchospasm
- iii. If monitoring device were to fail and Colorimetric ETCO2 Detector available:
 1. Inspect EZ Cap device to ensure purple color prior to application.
 2. Apply the EZ Cap to the tube after placement in a patient's mouth or nose.
 3. Ventilate with a Bag Valve device attached to the EZ Cap a minimum of six ventilations with 100% oxygen.
 4. Observe the color on the EZ Cap.
 - a. Purple should change to Gold if CO2 is present in the exhaled air. This is indicative of correct placement of the ETT in the trachea.
 - b. Absent or poor color change is suggestive of incorrect placement of the ETT or absence of circulation. Confirm tube placement by direct laryngoscopy, if possible, or remove the tube and re-attempt intubation. ("Purple with a pulse is poor...pull it out. Gold is good.")
- iv. Nasal Cannula with Capnometry
 1. Use in non-intubated patients with a GCS < 14 who receive sedation or analgesia.
 2. Patients who are a GCS > 8 but show signs of decreased work of breathing.
 3. Persistently low ETCO2 (sustained < 25 mmHg) is a sign of poor perfusion or metabolic acidosis associated with poor outcomes.

IV. DOCUMENTATION:

- a. EMS Charts

V. REFERENCES: None